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CSE 471

Assignment 1

Section 09

① a)

	2012	2013	2014	2015
Total benefits	0	2065500	2148120	2212564
Total costs	279000	461000	469700	483252
Net benefit = total benefit - total cost	(279000)	1604500	1678420	1729312
cumulative net benefits	(279000)	1325500	3003920	4733232

Total benefits = 0 + 2065500 + 2148120 + 2212564

= 6426184

Total costs = 279000 + 461000 + 469700 + 483252

= 1692952

$$ROI = \frac{\text{total benefits} - \text{total costs}}{\text{Total costs}}$$

$$= \frac{6426184 - 1692952}{1692952} \times 100$$

$$= 2.7958 \times 100$$

$$= 279.5\%$$

b) $$BEP = \text{number of years of neg cash flow} + \frac{\text{first positive cumulative net benefit years (net benefits - cumulative net bene)}}{\text{first positive cumulative net benefit years (net benefits)}}$$

$$= 0 + \frac{1604500 - 1325500}{1604500}$$

$$= 0.173 \text{ years}$$

c) $r = 6\% = 0.06$

$$PV = \frac{\text{cash flow}}{(1+r)^n}$$

	2012(0)	2013(1)	2014(2)	2015(3)	Total
Total benefit	0	2065500	2148120	2212564	
PV of total benefits	0	1948584	1911819	1857711	5718114
Total costs	279000	461000	469700	483252	
PV of total costs	279000	434905	418031	405747	1537683

$PV_0(TB) = 0$

$$PV_1(TB) = \frac{2065500}{(1+0.06)^1}$$

$= 1948584$

$$PV_2(TB) = \frac{2148120}{(1+0.06)^2}$$

$= 1911819$

$$PPV_3(TB) = \frac{2212564}{(1+0.06)^3}$$

$= 1857711$

$$PV_0(TC) = \frac{279000}{(1+0.06)^0}$$

$= 279000$

$$PV_1(TC) = \frac{461000}{(1+0.06)^1} = 434905$$

$\cancel{= 288125}$

$$PV_2(TC) = \frac{469700}{(1+0.06)^2} = 418031$$

$\cancel{= 183476}$

$$PV_3(TC) = \frac{483252}{(1+0.06)^3}$$

$\cancel{= 117981}$

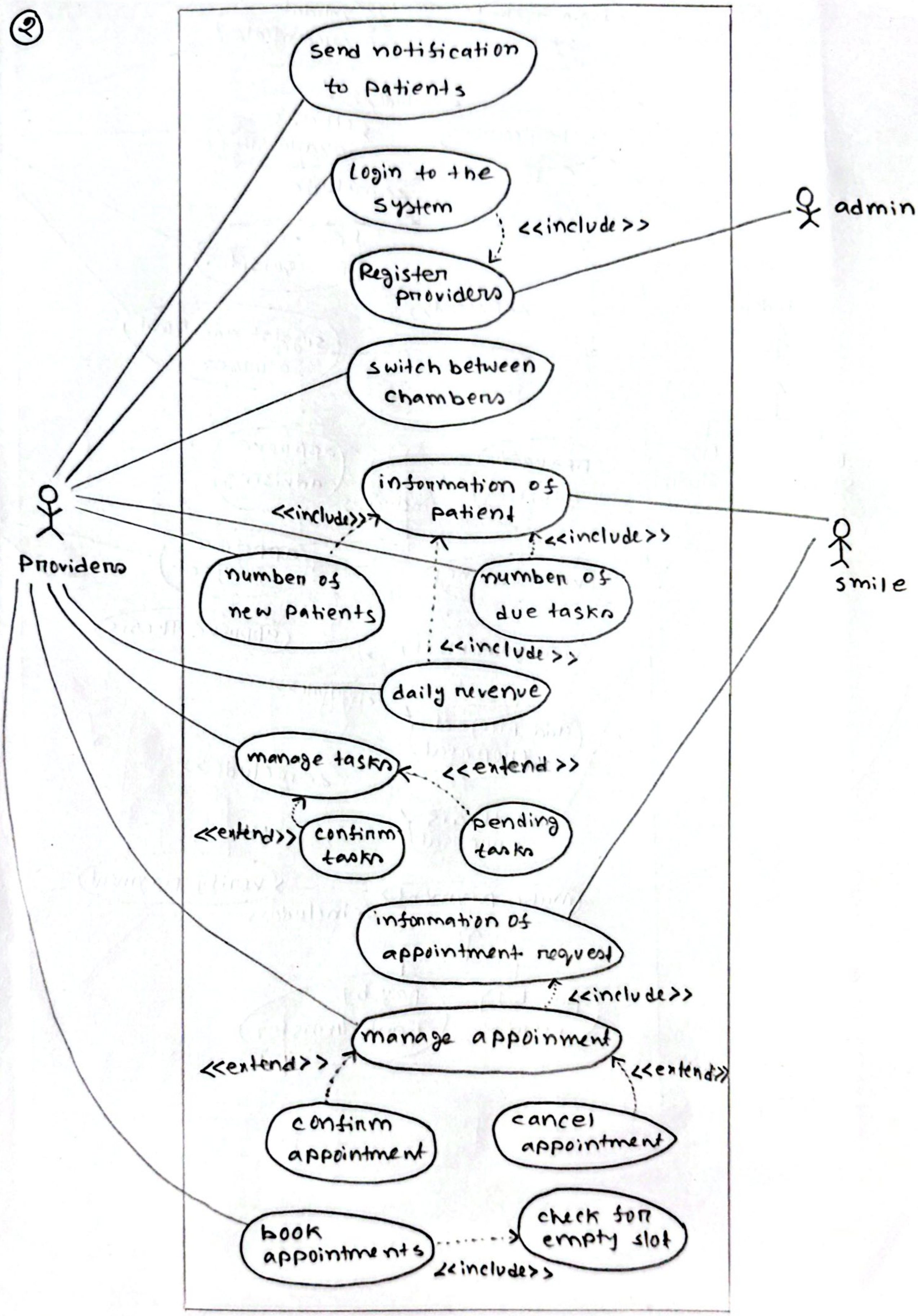
$= 405747$

NPV = ΣPV of total benefits $-$ ΣPV of total cost

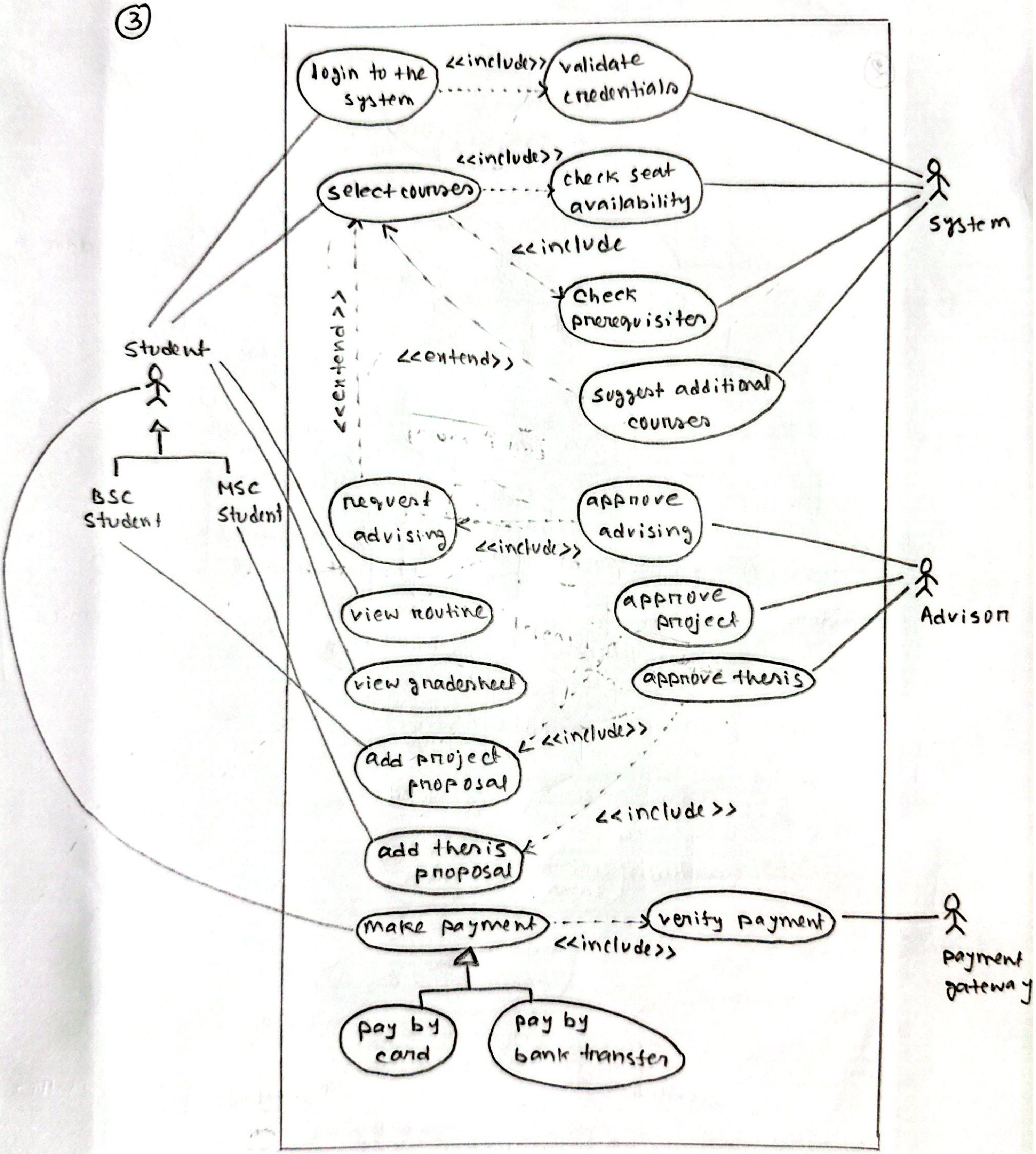
$= 5718114 - 1537683$

$= 4180431$

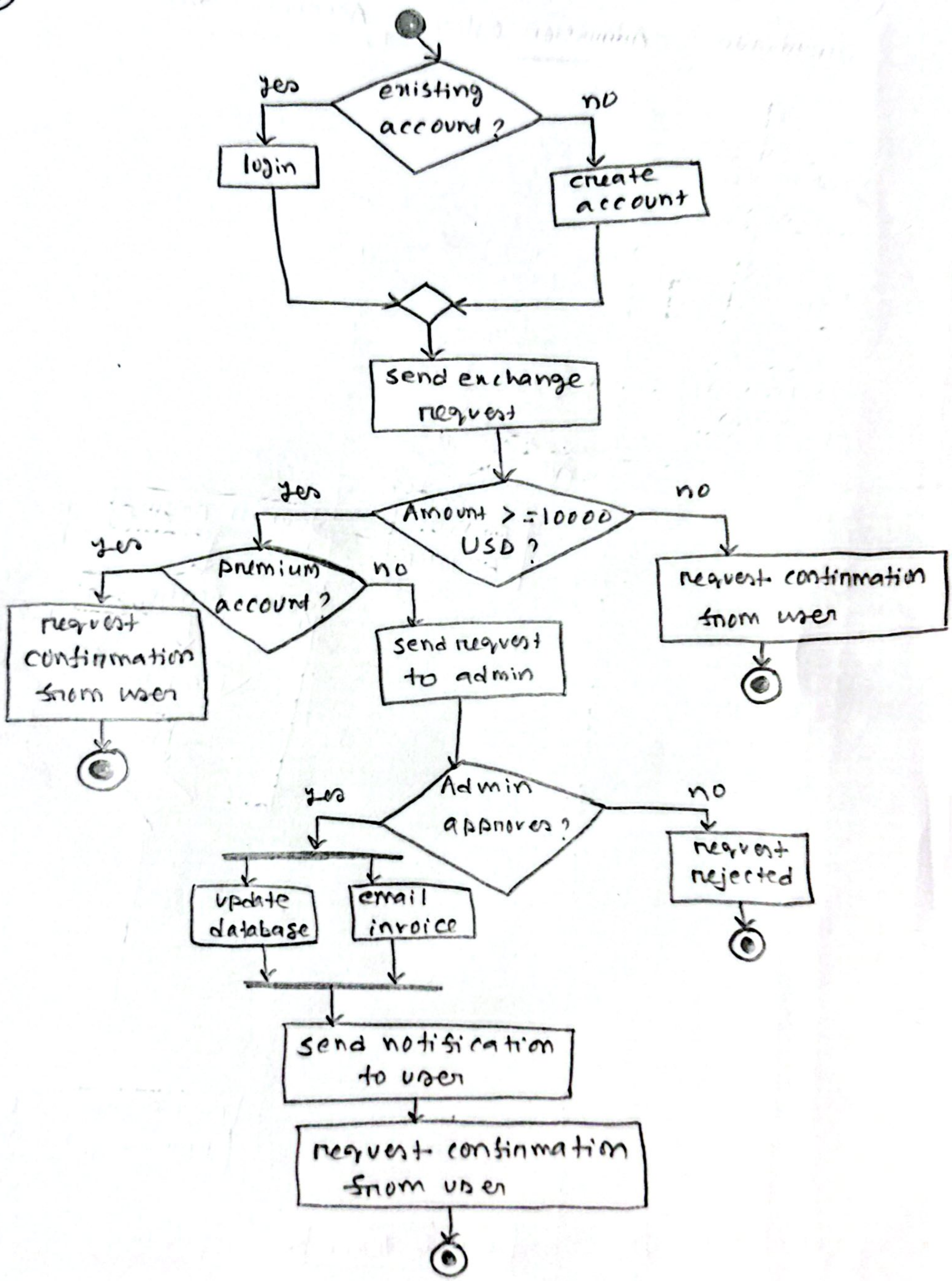
2


Send notification to patients
Login to the system
<<include>>
Register providers
admin
switch between chambers
information of patient
<<include>>
<<include>>
Providers
number of new patients
number of due tasks
smile
<<include>>
daily revenue
manage tasks
<<extend>>
<<extend>>
confirm tasks
pending tasks
information of appointment request
<<include>>
manage appoinment
<<extend>>
<<extend>>
confirm appointment
cancel appointment
book appointments
check for empty slot
<<include>>

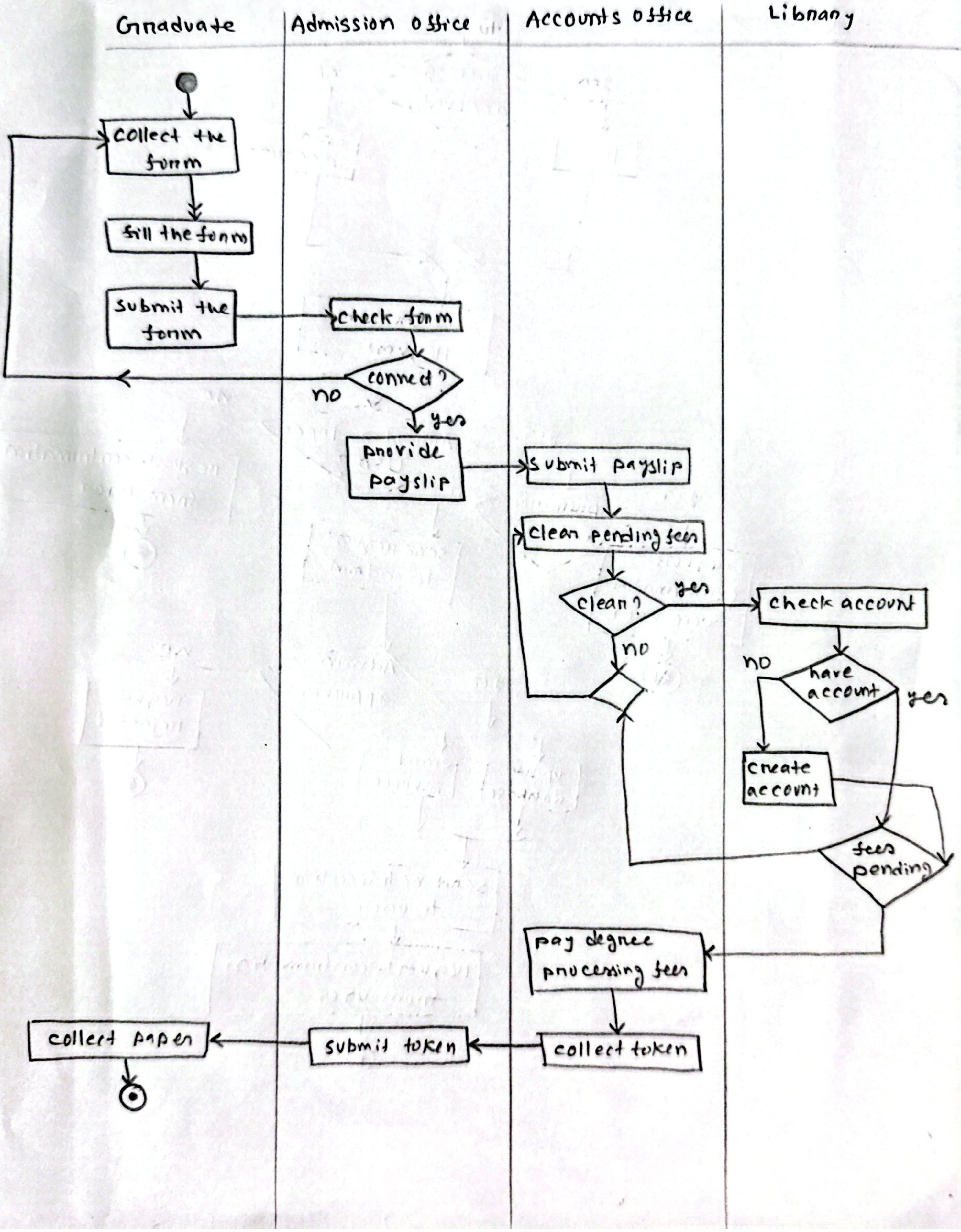
3


login to the system
<<include>>
validate credentials
select courses
<<include>>
check seat availability
<<include
Check prerequisites
<<extend>>
<<extend>>
suggest additional courses
System
Student
BSC Student
MSC Student
request advising
approve advising
<<include>>
Advisor
view routine
approve project
view gradesheet
approve thesis
add project proposal
<<include>>
<<include>>
add thesis proposal
make payment
verify payment
<<include>>
payment gateway
pay by card
pay by bank transfer

A


existing account ?
yes
no
login
create account
send exchange request
Amount >=10000 USD ?
yes
no
premium account ?
yes
no
request confirmation from user
send request to admin
request confirmation from user
Admin approves ?
yes
no
request rejected
update database
email invoice
send notification to user
request confirmation from user

5


Graduate
Admission office
Accounts office
Library
collect the form
fill the form
submit the form
check form
correct?
no
yes
provide payslip
submit payslip
clear pending fees
clear?
yes
no
check account
no
have account
yes
create account
fees pending
pay degree processing fees
collect token
submit token
collect paper